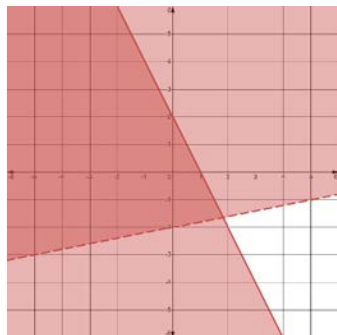


17 < 4 17 is not < 4.

Consider the system of inequalities,  $\begin{cases} y + 3 > 0.2(x + 5) \\ y - 4 \leq -2(x + 1) \end{cases}$   $\begin{cases} y > 0.2x - 2 \text{ or } y > \frac{1}{5}x - 2 \\ y \leq -2x + 2 \end{cases}$

6. Graph the boundary lines of the system.



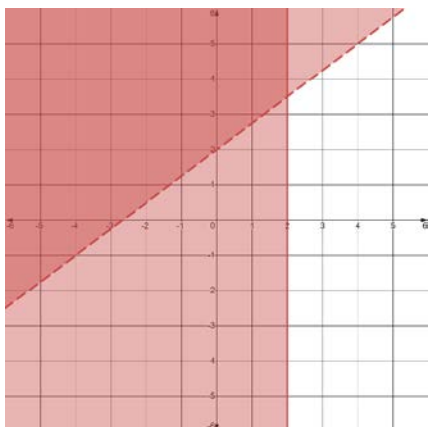
Commented [FAC1]: Link to graph:  
<https://www.desmos.com/calculator/b7vfzjvv5>

Use substitution to determine whether the given ordered pairs are solutions to the system.

|              | $y + 3 > 0.2(x + 5)$                          | $y - 4 \leq -2(x + 1)$                           | Solution to the system?                                          |
|--------------|-----------------------------------------------|--------------------------------------------------|------------------------------------------------------------------|
| 7. $(-3, 4)$ | $4 + 3 > 0.2(-3 + 5)$<br>$7 > 0.4 \checkmark$ | $4 - 4 \leq -2(-3 + 1)$<br>$0 \leq 4 \checkmark$ | <input type="radio"/> Yes<br><input type="radio"/> No            |
| 8. $(5, 2)$  | $2 + 3 > 0.2(5 + 5)$<br>$5 > 2 \checkmark$    | $2 - 4 \leq -2(5 + 1)$<br>$-2 \leq -12 \times$   | <input type="radio"/> Yes<br><input checked="" type="radio"/> No |

9. Complete the graph by shading the region included in the solution set of the system.

10. Graph the solution set of the system of inequalities:  $\begin{cases} x \leq 2 \\ y > \frac{3}{4}x + 2 \end{cases}$



Commented [FAC2]: Link to graph:  
<https://www.desmos.com/calculator/vozlb6myti>